

British schoolteacher and meteorologist John Dalton advanced the concept of atomic theory, which is a cornerstone of physics and chemistry. His discoveries about the nature of elements and atoms and the way they combine were essential to establishing chemistry as a modern science.

MILESTONES

WEATHER INSIGHTS

Begins a lifelong habit of keeping daily weather records, including pressure and wind speed, in 1787.

NATURE OF GASES

Observes that gases are affected by pressure and heat; in 1801, he forms his law of partial pressures.

DIFFERENCES IN ATOMS

Becomes the first person to calculate and make charts of relative atomic weights in 1803.

WIDELY HONORED

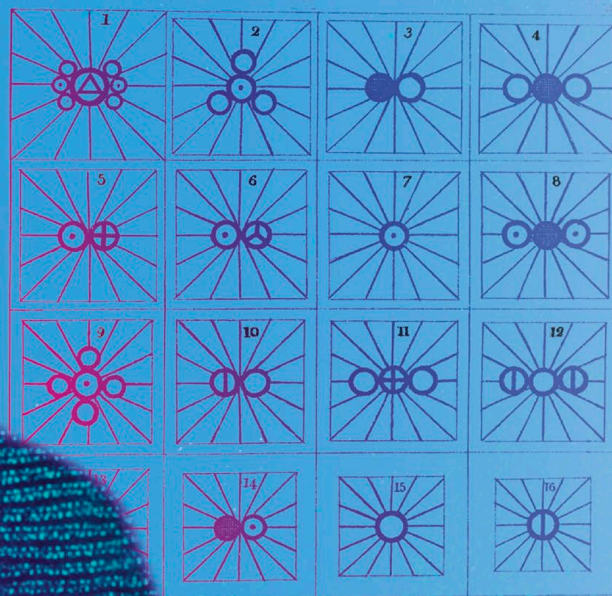
Receives Royal Society's medal in 1826. Elected to French and American Academies of Sciences.

Born into a Quaker family in Cumbria, England, John Dalton began work as a teacher aged 12. His first scientific paper described red-green color blindness; it is still sometimes called Daltonism. Both Dalton and his brother suffered from it, and he correctly believed it to be hereditary.

From a young age, Dalton kept records of rainfall, wind speeds, and air pressure. His meteorology textbook, published in 1793, was the first to correctly describe the hydrologic cycle (in which water evaporates from the oceans and falls as rain). Dalton also stated that water vapor did not combine chemically with air, as many thought, and that water's rate of evaporation depended on heat and wind speed.

Study of atmospheric gases

Dalton's interest in the weather led him to conduct experiments on the nature of atmospheric gases. In 1801, he gave a series of lectures in Manchester about his groundbreaking findings. These included the



Dalton's research into atmospheric gases led him to realize that the size and mass of atoms varies between elements. He made diagrams showing their structures and diameters.